FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS

PART TEST –III
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A, D	A, B, D	A
2.	A, C, D	A, B, C	B
3.	A, C	A, B, C	D
4.	A, B, C, D	A, B, C	C
5.	A, B, D	A, B, C	A, C
6.	A, B, C, D	D	A, C, D
7.	A, B, D	C, D	C
8.	A, B, C	A, B, C	A, B, C
9.	D	B	B
10.	A	A	A
11.	B	C	C
12.	B	C	C
13.	A	B	D
14.	C	A	D
15.	C	B	A
16.	A	B	D
17.	A	A	B
18.	B	C	D
19.	A	D	C
20.	C	A	C

Physics

PART – I

9. Use symmetry arguments.

10. zero, due to symmetry

11. B due to infinite wire $\propto I/\ell$

12. $\vec{B} = \vec{B}_1 + \vec{B}_2$

13-14. Use Kirchoff's Law

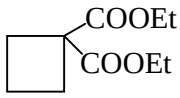
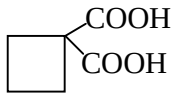
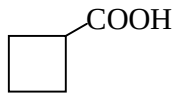
Chemistry

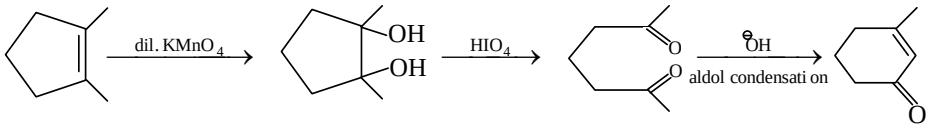
PART – II

SECTION – A

5. $(CH_3)_2CHCOOR \xrightarrow[(ii)\ H_3O^+]{(i)\ 2CH_3MgBr} (CH_3)_2CH-\underset{\displaystyle CH_3}{\overset{\displaystyle OH}{\overset{|}{\underset{|}{C}}}}-CH_3$

R = C_2H_5, C_6H_5

6. $CH_2(COOEt)_2 \xrightarrow[(ii)\ BrCH_2CH_2CH_2Br]{(i)\ NaOEt}$  $\xrightarrow{H_3O^+}$  $\xrightarrow{\Delta}$ 

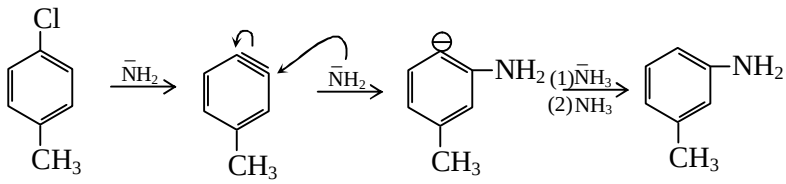
7. 

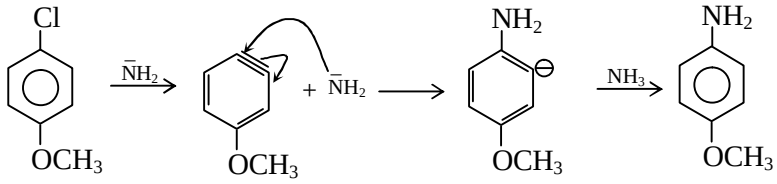
(A) (B) (C)

8. ABS is a polymer of acrylonitrile, butadiene and styrene.
cis – polyisoprene is a natural rubber.

13. Aldehyde, without α hydrogen, can respond to Cannizzaro reaction.

14. Ph CO CHO contains one-CO-and another –CHO group adjacently without having α–hydrogen.

15. 

16. 

Mathematics PART – III

1. $x = 1; y = 2$
$2 = a + b + c$ ..... (1)
$x = 0, y = 0 \Rightarrow c = 0 \Rightarrow a + b = 2$
Now, $\left.\frac{dy}{dx}\right|_{(0,0)} = 2ax + b = 1$
$\therefore b = 1, a = 1$
Hence, the curve is $y = x^2 + x$
$A = \int_{-1/2}^{0} \left(x^2 + x - x\right)dx = \int_{-1/2}^{0} \left(x^2\right)dx = \frac{1}{24}$ sq. units

2. $I = \int \sqrt{1 + 2\,\text{cosec}\,x\cot x + 2\cot^2 x}\,dx$
$= \int \sqrt{\text{cosec}^2 x + 2\,\text{cosec}\,x\cot x + \cot^2 x}\,dx$
$= \int \text{cosec}\,x + \cot x\,dx$

3. Degree of $f(x) = n$
Degree of $f'(x) = n - 1$
Degree of $f''(x) = n - 2$
$\therefore n = (n - 1) + (n - 2) = 2n - 3$
$\therefore n = 3$
Hence, $f(x) = ax^3 + bx^2 + cx + d, (a \neq 0)$
$f'(x) = 3ax^2 + 2bx + c$; $f''(x) = 6ax + 2b$
$a = \frac{1}{18}$

4. For continuity of f at $x = 0$, we have
$k = f(0) = \lim_{x\to 0} f(x) = 1 + 3\lim_{x\to 0} \frac{\sec x - 1}{3x^2}$ using L.H. Rule
$f(0) = 1 + \frac{1}{2} = \frac{3}{2}$

5. $f(x) = (x^2 + bx + c)e^x$
$\therefore f'(x) = (x^2 + (b + 2)x + (b + c))e^x$
$f(x) > 0$ iff $D = b^2 - 4ac < 0$
Now, $f'(x) > 0$ iff $D' = (b + 2)^2 - 4(b + c) = D + 4 < 0$
Thus for $f'(x) > 0$ $D + 4 < 0$ holds
$\Rightarrow D < 0 \Rightarrow f(x) > 0$

6. $y = f(x) = \frac{x^2 - 1}{x^2 + 1} = 1 - \frac{2}{x^2 + 1}$
$f'(x) = \frac{4x}{\left(x^2 + 1\right)^2}$
$x > 0$, f is increasing
$x < 0$, f is decreasing
range is $[-1, 1] \Rightarrow$ into (A – false) minimum value occurs at $x = 0$
$f(0) = -1$ (C – false)
$A = 2\int_0^{\infty}\left(1 - \frac{x^2 - 1}{x^2 + 1}\right)dx = 2\pi$ (D – false)

7. $D^* f(x) = 2f(x) f'(x)$
$D^* (x \ln x) = 2x \ln x (1 + \ln x)$

9. Given $f(x)\cdot f'(-x) = f(-x)\cdot f'(x)$
$$\frac{f'(x)}{f(x)} = \frac{f'(-x)}{f(-x)}$$
Integrating $\ln f(x) = -\ln f(-x) + c$
$\ln (f(x)\cdot f(-x)) = c$
$f(x)\cdot f(-x) = c$
$f(0) = 3 \therefore f^2(0) = c \Rightarrow c = 9$
$\therefore f(x)\cdot f(-x) = 9$

10. Let $I = \int_{-51}^{51} \frac{dx}{3 + f(x)}$
$$I = \int_{-51}^{51} \frac{dx}{3 + f(-x)} \text{ (Using king)}$$
$\therefore I = 17$

11. $y = e^t \sin t \Rightarrow \frac{dy}{dt} = e^t(\cos t + \sin t)$
$x = e^t \cos t \Rightarrow \frac{dx}{dt} = e^t[\cos t - \sin t]$
$$\therefore \frac{dy}{dx} = \frac{\cos t + \sin t}{\cos t - \sin t} = \tan\alpha$$
$$\therefore \tan\left(\frac{\pi}{4} + t\right) = \tan\alpha$$
$$\Rightarrow \frac{\pi}{4} + t = \alpha$$
$$t = \alpha - \frac{\pi}{4}$$

12. $f(t) = \int e^t(\cos t + \sin t)dt = e^t \sin t + c$
$$\therefore f\left(\frac{\pi}{2}\right) - f(0) = \left(e^{\pi/2} + c\right) - c = e^{\pi/2}$$

13. $x = f(t) = a^{\ell n(b^t)} = a^{t\ell nb}$ ..... (1)
$y = g(t) = b^{-\ell n(a^t)} = \left(b^{\ell na}\right)^{-t} = \left(a^{\ell nb}\right)^{-t} = a^{-t\ell nb}$
$\therefore y = g(t) = a^{\ell n(b^{-t})} = f(-t)$ ..... (2)
From equation (1) and (2) we get, $xy = 1$

14. $f(t) = g(t) \Rightarrow f(t) = f(-t) \Rightarrow t = 0$
{$\because$ f(t) is one-one function}
At $t = 0$; $x = y = 1$
$\because xy = 1$
$$\therefore \frac{dy}{dx} = -\frac{1}{x^2} \text{ and } \frac{d^2y}{dx^2} = \frac{2}{x^3}$$
At $x = 1$, $\frac{d^2y}{dx^2} = 2$

15.-16. Since, minimum value is zero
Hence, y = f(x) touches the x-axis and mouth opening upward i.e. a > 0
Given f(x – 4) = f(2 – x) (x → x + 3)
f(x – 1) = f(–1 – x)
f(–1 + x) = f(–1 – x)
Hence, f is symmetric about the line x = –1
$\therefore$ $f(x) = a(x+1)^2$
Now, also given $f(x) \geq x \ \forall \ x \in R$
$\therefore$ $f(1) \geq 1$ ..... (1)

and $f(x) \leq \left(\frac{x+1}{2}\right)^2$ in (0, 2)

$f(1) \leq 1$ ..... (2)
From (1) and (2), we get f(1) = 1
Now $f(x) = a(x+1)^2$

$f(1) = 4a = 1 \Rightarrow a = \frac{1}{4}$

$\therefore$ $f(x) = \frac{(x+1)^2}{4}$ (Now proceed)

17. (P) Let $I_1 = \int \frac{x^2\left(x^6+x^5-1\right)}{\left(2x^6+3x^5+2\right)^2}dx = \int \frac{x^2\left(x^6+x^5-1\right)}{x^6\left(2x^3+3x^2+\frac{2}{x^3}\right)^2}dx = \int \frac{\left(x^2+x-x^{-4}\right)}{\left(2x^3+3x^2+2x^{-3}\right)^2}dx$

Putting $2x^3 + 3x^2 + 2x^{-x} = t$

$\therefore$ $\frac{x^6+x^5-1}{x^4}dx = \frac{1}{6}dt$

Hence, $I_1 = \frac{1}{6}\int\frac{dt}{t^2} = \frac{-1}{6t} + c$

(Q) Putting $4x^5 + 5x^4 + 10x^2 = t$
$20(x^4 + x^3 + x)dx = dt$

(R) Putting $1 + x^{-2} + x^{-5} = t$
$(-2x^{-3} - 5x^{-6})dx = dt$

(S) $\int \frac{2x+1}{x^3\left(1+\frac{4}{x}+\frac{1}{x^2}\right)^{3/2}}dx = \int \frac{2x^{-2}+x^{-3}}{\left(1+\frac{4}{x}+\frac{1}{x^2}\right)^{3/2}}dx$

Now, put $\frac{1}{x^2}+\frac{4}{x}+1 = t^2$

18. Given $I = \int \frac{e^x}{e^{4x}+1}dx$ and $J = \int \frac{e^{-x}}{e^{-4x}+1}dx$

$J = \int \frac{e^{4x}\cdot e^{-x}}{1+e^{4x}}dx = \int \frac{e^{3x}}{1+e^{4x}}dx$

$\therefore$ $I + J = \int \frac{e^x\left(1+e^{2x}\right)}{1+e^{4x}}dx$

Put $e^x = t \Rightarrow e^x dx = dt$

$\therefore$ $I + J = \int \frac{1+t^2}{1+t^4}dx = \int \frac{1+\frac{1}{t^2}}{\left(t-\frac{1}{t}\right)^2+2}dt$

Put $t-\frac{1}{t}=y \Rightarrow dy=\left(t+\frac{1}{t^2}\right)dt$

$\therefore I+J=\int\frac{dy}{y^2+2}=\frac{1}{\sqrt{2}}\tan^{-1}\frac{y}{\sqrt{2}}=\frac{1}{\sqrt{2}}\tan^{-1}\frac{t^2-1}{\sqrt{2}t}$

$I+J=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right)+c$ ..... (1)

Similarly, $J-I=\frac{1}{2\sqrt{2}}\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}$ ..... (2)

Equation (1) – (2) gives $I=\frac{1}{2\sqrt{2}}\left[\tan^{-1}\frac{e^{2x}-1}{\sqrt{2}e^x}-\frac{1}{2}\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right]$

Equation (1) + (2) gives $J=\frac{1}{2\sqrt{2}}\left[\tan^{-1}\frac{e^{2x}-1}{\sqrt{2}e^x}+\frac{1}{2}\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right]$

19. (P) $f'(x)=\frac{g'(x)}{\sqrt{1+g^3(x)}}$ and $g'(x) = [1 + \sin(\cos^2 x)](-\sin x)$

Hence, $f'(x)=\frac{\left[1-\sin\left(\cos^2 x\right)(\sin x)\right]}{\sqrt{1+g^3(x)}}$

$f'\left(\frac{\pi}{2}\right)=\frac{1+0}{\sqrt{1+g^3\left(\frac{\pi}{2}\right)}}=\frac{-1}{1+0}=-1$

(R) Maximum when $a = -1, b = 2$

$\Rightarrow a + b = 1$

(S) If $\lim\limits_{x\to0}\frac{\sin 2x}{x^3}+a+\frac{b}{x^2}=0$

$\Rightarrow \lim\limits_{x\to0}\frac{\sin 2x+ax^3+bx}{x^3}=0$ for limit to be exist $2 + b = 0$

$\therefore b = -2$

$\therefore \lim\limits_{x\to0}\frac{\sin 2x+ax^3-2x}{x^3}=0$

Applying L.H. rule

$\lim\limits_{x\to0}\frac{2\cos 2x+3ax^2-2}{3x^2}=0$

$\therefore a=\lim\limits_{x\to0}\frac{2(1-\cos 2x)}{3x^2}$

$\Rightarrow a=\frac{4\sin^2 x}{3x^2}=\frac{4}{3}$

20. (P) $I_1=\int_2^3\left((x-1)^3+(4-x)^3+x\right)\cos\pi x dx$ ..... (1)

Applying king property we get $I_1=\int_2^3\left((4-x)^3+(x-1)^3+5-x\right)\cos\left(\pi(5-x)\right)dx$

$$I_1 = -\int_2^3 \left((4-x)^3 + (x-1)^3 + 5 - x\right)\cos \pi x dx \qquad \text{..... (2)}$$

Now, (1) + (2), gives $2I_1 = \int_2^3 (2x-5)\cos \pi x dx$

$$\Rightarrow 2I_1 = \int_2^3 2x\cos \pi x dx - \int_2^3 5\cos \pi x dx$$

$$2I_1 = 2\int_2^3 x\cos \pi x \, dx \; x - 5\int_2^3 \cos \pi x \, dx$$

$$2I_1 = 2\left(\frac{x\sin \pi x}{\pi} - \int \frac{\sin \pi x}{\pi} dx\right)_2^3 - 5\left(\frac{\sin \pi x}{\pi}\right)_2^3 = 0 + \frac{1}{\pi^2}(-1-1) = -\frac{2}{\pi^2}$$

$\Rightarrow 50\pi^2 I_1 = -100$

Hence, $|50\ \pi^2 I_1| = 100$